

A 3D Multiphase Continuum Model for Atoms

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September 25, 2026

*Mathematical formulation and the p5.js templates by the author;
developed from them with AI assistance (Claude, Anthropic).*

Abstract

RealQM is a model of atoms and molecules in the form of free-boundary multiphase continuum mechanics in three space dimensions, based on non-overlapping unit electron charge densities meeting with continuity at free interfaces and interacting by Coulomb potentials without self-repulsion and with kernel potentials. The ground state is computed as the steady state of a *parabolic relaxation*, which decreases the Coulomb energy while maintaining continuity of charge density across the interfaces, their position being an unknown of the same dynamics. The model is parameter-free: its inputs are the nuclear charge and the number of electrons, and the energy minimised is the Coulomb energy alone, with no constant adjusted anywhere. Work scales as N times the mesh, N being the number of electrons, so the cost is linear in N at fixed resolution; RealQM applied to molecules, to condensed phases and to nuclei is presented in companion articles.

We here let RealQM derive a *periodic table from scratch* — the ground states of the neutral atoms, nothing fitted — and compare it with the textbook table. The shell structure, meaning how many charges occupy each shell, is *not* a result of energy minimisation *only*: left free to choose the packing the energy is monotone in it, its minimiser absorbing the core into the valence, wrong in kind and worsening with Z . The structure comes instead from the build-up, each new charge continuing the outermost shell or, that shell being complete, starting the next, a filled shell being more rigid than the charge arriving at it. Given the structure, the same energy is accurate to one or two per cent across the first three periods. The rule recovers Li, Be, B and C with no empirical input, and we report it as established to carbon and open beyond.

The capacities follow from the packings themselves. The octet appears as 4+4, two dual tetrahedra forming a cube, never as a single shell of eight, which over-binds by 9–22% — and not as the 2+6 of the orbital picture: the same eight, decomposed differently, and the decomposition is what bonding geometry depends on; and composing rows from that octet together with the antipodal pair gives the row lengths 2, 8, 8, 18, 18 and the closures 2, 10, 18, 36, 54, the fourth row being 2+8+8 rather than 9+9. The account is exact through xenon and open at the sixth row. We do not claim the familiar $2n^2$ as a result of this model, that count being the hydrogenic degeneracy. Computed directly from the shell energies, the Allen configuration energy reproduces the electronegativity ordering across period 2 exactly, with a gradient too steep by a factor of about two.

Results are computed in spherical symmetry, a specialisation available because atoms are spherical; the three-dimensional form displays qualitative features of the solution — a bond, and an electron arriving at a closed core wrapping around it rather than entering. Every atom is solved all-electron, with no core radius and no fitted length. From the fourth row the shells interpenetrate, the $3d$ lying inside the $4s$ that fills after it, so that the build-up ceases to reach

the configuration from $Z \approx 21$, which bounds the calculation at argon. Ionization and excitation are treated elsewhere.

Keywords: atomic structure; real-space methods; multiphase wave functions; free-boundary problems; electron packing; Aufbau principle; electronegativity; imaginary-time propagation.

1 Introduction

RealQM is a model of atoms and molecules in the form of free-boundary multiphase continuum mechanics in three space dimensions, based on non-overlapping unit electron charge densities meeting with continuity at free interfaces and interacting by Coulomb potentials without self-repulsion and with kernel potentials. The ground state is computed as the steady state of a *parabolic relaxation*, or imaginary time stepping, which decreases the Coulomb energy while maintaining continuity of charge density across the interfaces, their position being an unknown of the same dynamics. What is reached is the equilibrium accessible from the state one starts in, so the minimisation is *path dependent* — ordinarily an invisible distinction, since that equilibrium is the minimum, but the one that decides the shell structure of atoms. The model is parameter-free: its inputs are the nuclear charge and the number of electrons, and the energy minimised is the Coulomb energy alone, with no constant adjusted anywhere. Computational work scales as N times the mesh, N being the number of electrons, since each density requires one Poisson solve on a grid in three space dimensions; the cost is thus linear in N at fixed resolution, which allows the simulation of large molecules and, with successive model reduction, opens to clusters of molecules; RealQM applied to molecules, to condensed phases and to nuclei is presented in companion articles [1].

We here take the different route, to *maximal simplicity*: the model entire, in its smallest implementation and on the smallest systems, with nothing fitted and nothing carried over from the conventional account.

In an atom the densities arrange themselves about the kernel in *shells*: an inner region of tightly bound charge, then successive outer layers, each shell holding a definite number of densities and separated from the next by a free interface. This shell structure — how many charges occupy each shell — is *not* a result of energy minimisation *only*. It comes from the packing, electron by electron, as the atom is built: a dynamic Aufbau in which each new charge either continues to fill the outermost shell or, if that shell is complete, starts the next. A filled shell is shut to it: the charge in a complete shell is a more rigid arrangement than that of an arriving electron, and resists its inclusion. Given the structure, the minimisation then settles the interfaces and the energy (Section 5.2).

An atom is spherically symmetric and its shells are concentric. The model therefore admits a reduction to a single radial dimension, and it is in that form that the table is computed here. The reduction is a specialisation of the same Coulomb energy, not a second model; but radial averaging keeps only the number of charges in a shell and discards their angular arrangement, which is where the packing lives, so it is the three-dimensional form that decides a packing question (Section 3).

Codes. We consider here a *minimal implementation* — minimal in the code, not in the model, which is carried in full — essentially three lines expressing the update of the charge densities, the update of the potentials and the update of the free boundary, iterated to a joint steady state (Section 4). It is used in two forms: spherically symmetric in one dimension, some 175 lines, covering all Z ; and fully three-dimensional, some 300 lines, used at small Z (Section 10). The three-dimensional form is the general one, and the one that extends to molecules. Both are short

enough to be read end to end, so the claim that nothing is fitted can be checked rather than taken on trust.

Objective. Let RealQM compute a periodic table from scratch: the *neutral ground states*, and for each of them the *total energy*, the *shell structure*, and the *Aufbau* by which that structure forms. The result is compared against the high-accuracy atomic data assembled from specialised treatments [8, 4, 6, 5, 7, 3, 9, 2], each suited to its own range; RealQM does not attempt that kind of accuracy, and the test is of a different kind: one parameter-free model, based on Coulomb physics, applied uniformly to every element and asked for the gross structure of the whole table at once. Every atom is solved all-electron, with the bare nuclear term throughout and no fitted length anywhere. What the heavier elements require is not a parameter but resolution: the innermost shell contracts as $1/Z$, so the mesh is refined where it must be, and nothing else changes. What is computed throughout is the *ground states*, which is what a periodic table is about. Ionization and excitation are treated elsewhere, as are molecules, condensed phases of thousands of atoms, and nuclei, with larger implementations of the same three updates [1].

Comparison with textbook quantum mechanics. The two rest on the same ingredients — kinetic energy and Coulomb interaction, and nothing else — and differ in the form taken by the wave function. Textbook quantum mechanics uses a single global wave function over a $3N$ -dimensional configuration space, and the Schrödinger equation for it is *linear*. RealQM uses a sum of one-electron wave functions with local, non-overlapping supports in three-dimensional physical space, the supports themselves being unknowns, which makes it a *nonlinear* free-boundary problem. The physics put in is the same; the form of the description is not, and the difference is worth stating plainly rather than as a verdict.

Density functional theory is the nearer comparison, since it too works in three dimensions. The difference is that in RealQM the electrons retain their individuality, by occupying separate regions of space, and in density functional theory they do not, the state being a single common density.

Exclusion is realised differently in each — by antisymmetry of a function on configuration space, or by non-overlap of domains in three-space — and neither is shown here to be an approximation to the other. What is compared in this paper is what each produces for the periodic table. The reader should take the wider ontological reading advanced elsewhere [1] as speculative, pending evidence stronger than atomic energetics can supply; nothing here depends on it.

2 The multiphase atom model

RealQM models Coulomb interaction between non-overlapping unit electron charge densities and point-like kernels, posed as a multiphase free-boundary problem of continuum mechanics. Each electron is one unit of charge spread as a density over a domain of three-dimensional space; the domains tile space without overlapping; the densities meet with continuity on a free boundary, its position determined by the solution rather than imposed; and the only interaction is the ordinary Coulomb one, between the densities themselves and between each density and the kernels: the kinetic energy of the densities and their Coulomb energy are the whole of the functional. We summarise below only what is needed here; derivations are in [1]. For an atom of nuclear charge Z the N -electron wave function $\Psi \in H^1(\mathbb{R}^3)$ has the form

$$\Psi(\mathbf{x}) = \sum_{i=1}^N \psi_i(\mathbf{x}), \quad \psi_i \in H^1(\Omega_i), \quad \int_{\Omega_i} \psi_i^2 d\mathbf{x} = 1, \quad (1)$$

on a partition of $\mathbb{R}^3$ into non-overlapping domains Ω_i with free boundaries $\Gamma_i = \partial\Omega_i$. Hartree atomic units are used throughout. The total energy is the standard Coulomb functional

$$TE(\Psi) = \underbrace{\frac{1}{2} \sum_i \int_{\Omega_i} |\nabla \psi_i|^2}_{\text{kinetic}} + \underbrace{\int_{\mathbb{R}^3} W \Psi^2}_{\text{kernel-electron}} + \underbrace{\sum_i \int_{\Omega_i} \sum_{j \neq i} V_j \psi_i^2}_{\text{electron-electron}}, \quad (2)$$

with kernel potential $W(\mathbf{x}) = -Z/|\mathbf{x}|$ and per-electron Hartree potentials $V_j(\mathbf{x}) = \frac{1}{2} \int_{\Omega_j} \psi_j^2(\mathbf{y})/|\mathbf{x} - \mathbf{y}| d\mathbf{y}$; self-repulsion is excluded by construction ($j \neq i$ on disjoint domains).

What holds a density against its kernel is then the first term. For a single electron the last term is empty and (2) reduces to the kinetic energy against the kernel attraction alone, $-\frac{1}{2}\Delta$ against $-Z/r$: that balance, and nothing else, fixes the size of the density, and for $Z = 1$ it is the hydrogen problem. A functional built instead on one common density $\rho = \sum_i \psi_i^2$ integrates ρ against itself, so the same balance carries a repulsion of the electron with its own field and the density comes out too diffuse; removing that term afterwards is what a self-interaction correction is for. Here there is nothing to remove.

Stationarity in ψ_i at fixed Ω_i gives the one-electron eigenproblem

$$\mathcal{H}_i \psi_i \equiv \left(-\frac{1}{2}\Delta + W + 2 \sum_{j \neq i} V_j \right) \psi_i = E_i \psi_i \quad \text{in } \Omega_i, \quad \frac{\partial \psi_i}{\partial n} = 0 \quad \text{on } \Gamma_i, \quad (3)$$

with continuity of ψ_i across the inter-electron interface and the homogeneous Neumann condition together constituting the **Bernoulli free-boundary condition**. Spatial exclusion plays the role of the Pauli principle: the electrons occupy disjoint territories rather than antisymmetrised orbitals.

For an atom the model reduces to spherical symmetry — a specialisation available because atoms are spherical — with Ψ a function of r alone, the domains Ω_i concentric shells, the free interfaces Γ_i radii R_s , and the same-shell self-term of a shell holding N_s charges reduced by $(N_s-1)/N_s$ (Section 3); the reduced equations are exhibited by the code of Section 10. Every atom reported here is solved all-electron, the nuclear term the bare $-Z/r$ throughout, with no core radius, no screened kernel and no fitted length anywhere.

3 What the spherical reduction costs

What the $(N_s-1)/N_s$ factor is, and what the reduction costs. The factor is not a correction added to the model: it is forced by the reduction. In 3D the N_s electrons of a shell occupy N_s separate one-electron domains and the $i \neq j$ rule counts $N_s(N_s-1)/2$ pairs. Smearing them into one spherical density of charge N_s makes the Hartree self-energy count N_s^2 pairs, of which N_s are spurious self-pairs; multiplying by $(N_s-1)/N_s$ removes exactly those. Nothing is fitted, and the parameter-free character of the model is untouched.

Smearing does a second thing, however, and this one is a genuine loss. It replaces the actual pair *separations* by their spherical average. The charges of a shell sit as far apart as the shell allows — antipodal for $N_s = 2$, trigonal for 3, tetrahedral for 4 — and those arrangements repel less than the angular average. For unit charges at a common radius R (units of $1/R$):

N_s	homogenised, $q = N_s$	$\times (N_s-1)/N_s$	packed	ratio
2	2.000	1.000	0.500	2.00
3	4.500	3.000	1.732	1.73
4	8.000	6.000	3.674	1.63

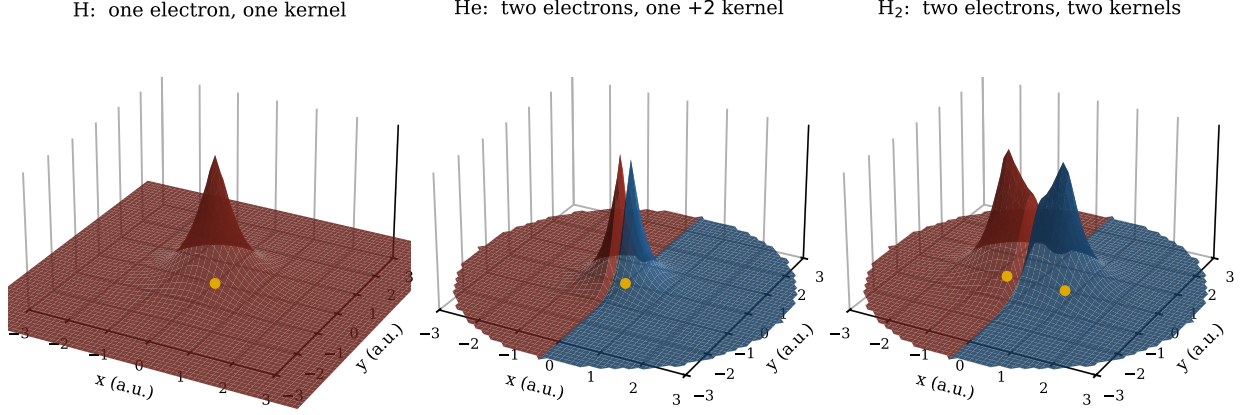


Figure 1: The model in three dimensions: charge density as surface elevation over the plane through the kernels, one colour per electron domain, kernels marked in yellow on the base plane. **H**: a single domain. **He**: two one-electron domains on one +2 kernel, meeting at a free interface through the nucleus. **H₂**: the same two domains with the kernels drawn apart to the bond length. He and H₂ differ only in where the kernels sit, which is this model’s account of what a bond is. The quantity drawn is $|\psi_i|^2$ on the territory domain i holds, unowned cells left unpainted, so the interface appears as the ridge between the two surfaces rather than as overlapping clouds. Cartesian grid, 120^3 over a box of 8 a.u., $\varepsilon = h$. The figure is of the partition; total energies are in Table 1.

So the reduced homogenised shell has the pair *count* right and the pair *geometry* wrong, overestimating intra-shell repulsion by a factor 1.6–2. Most of that is absorbed by the shell radius, which is free and expands until the energy re-minimises; what survives appears as a systematic under-binding, and it is largest at $N_s = 2$, where the ratio is largest. Helium is the clean test — one shell, two electrons, no core, no r_c , and an exact reference — and it under-binds by 2.0% (Table 1), with the sign the table predicts.

The thing discarded is precisely the angular correlation, which is what packing *means*. This is why the radial reduction cannot adjudicate a packing question on energetic grounds — it has no angles with which to tell 4+4 from a single 8-shell — and why the capacities of Section 5.1 are established by the counting argument and the 3D solve rather than by the radial energies.

4 Algorithm: the three-line solver

The system is solved by three coupled relaxations to a joint steady state — the *three-line solver*. We state them for the general case of Section 2; the spherical reduction stated there is the special case in which the domains are concentric shells and each ψ_i depends on r alone. At each step, for every domain Ω_i :

1. **Imaginary time stepping** of the amplitude — parabolic relaxation — $\partial_\tau \psi_i = -\mathcal{H}_i \psi_i$ with $\mathcal{H}_i$ from (3), relaxing ψ_i toward the lowest eigenfunction of $\mathcal{H}_i$; τ is an artificial time carrying the relaxation and has no physical meaning, only the steady state being used;
2. a **Poisson solve** for the per-electron potential V_i , which in spherical symmetry is the shell-theorem integral over the other shells;
3. a **boundary move**: each interface Γ_i is shifted in the direction that *decreases the jump* of

Ψ across it, driving it toward the continuity that $\Psi \in H^1$ demands — the Bernoulli free-boundary condition.

After each step ψ_i is renormalised; at convergence the boundaries are stationary and the per-electron energy is the Rayleigh quotient, supplying the per-shell energies. In the spherical case the scheme is discretised on a uniform grid $r_i = ih$ with an explicit step $\Delta\tau = \frac{1}{2}h^2$ (first order, cusp-limited; Section 7), and each of the three is a single one-dimensional grid sweep on $r_i = ih$, so every atom converges in a fraction of a second and the whole periodic table in about a minute on a laptop.

The algorithm is a relaxation, not a global minimisation. This is worth stating at the outset because it governs how the results below should be read. None of the three updates searches the space of configurations; each is a parabolic relaxation — the amplitude down the gradient of H_s , the potential toward its Poisson source, the interface toward continuity — and what they reach together is a *steady state*, a configuration at which all three forces balance. Such a state is a critical point of the Coulomb functional, and the ground state is one; but the relaxation arrives at whichever equilibrium lies at the end of the path it is on, and there is no step at which a lower configuration elsewhere in the space is offered and refused. Energy minimisation here is therefore *path dependent*: what is computed is the equilibrium reachable from the initial state, not the infimum over all states.

Ordinarily this distinction is invisible, because the reachable equilibrium *is* the minimum. It becomes the whole point in Section 5.2, where a lower configuration exists, is never reached, and is not the physical one either — so that the path, and not the infimum, is what selects the structure of the atom.

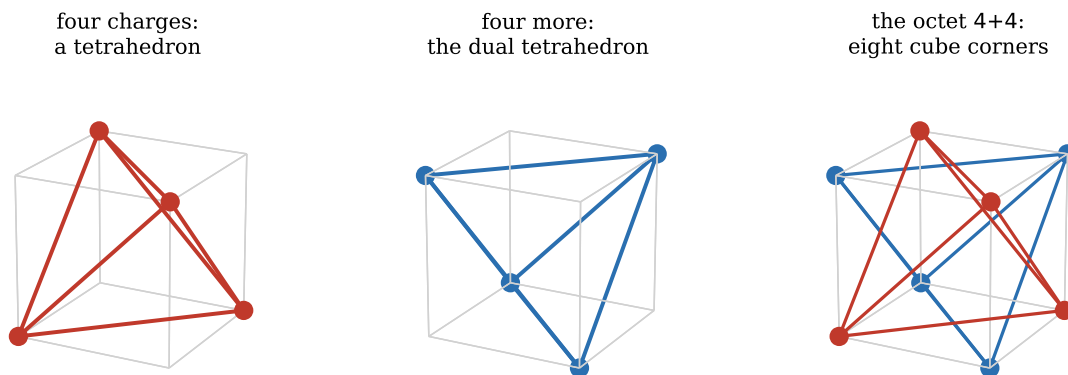
5 Results: total energies, packing, and electronegativity

We report three results. The electron packing (Section 5.1) and the all-electron total energies of the first three periods (Section 5.3) come from the spherically-reduced (1D radial) solver applied to every electron, with the bare nuclear charge and no fitted length. The electronegativities (Section 5.5) follow from the same shell energies. Periods 1–3 are covered in full, hydrogen through argon; beyond argon the shells cease to nest and the radial reduction fails (Section 5.4).

5.1 Electron packing: 4+4, not 8

The shell packing of each atom is supplied by the build-up of Section 5.2, not chosen by fitting energies. Two findings are robust across $Z = 2$ –18. *First*, comparing shells of capacity 2, 4, 6, 8, the **cap-4** packing matches reference total energies best everywhere: a capacity-8 shell over-binds by +9% to +22% (worsening with Z), and a closed octet is realised only as **4+4**, never as a single 8-shell — the geometrically expected result, since four unit charges pack as a stable tetrahedron while eight on one shell do not. *Second*, distributing each period’s electrons *evenly* across its cap-4 shells (balanced filling) rather than greedily reproduces the classic configurations 2+2+1 (B), 2+2+2 (C), 2+3+2 (N), 2+3+3 (O) and tightens the energies (e.g. C: +4.4% greedy $\rightarrow$ −1.1% balanced).

The organising principle is geometric: each electron is a localised cloud whose *size* is set by its distance from the kernel — inner electrons compact, outer ones diffuse — and the shells are the close-packings (capacities 2 and 4) of these clouds. Different packings of the same atom have different energies, and — importantly — the *unconstrained* energy minimum is *not* the physical configuration: in the 1D solver it favours the over-packed capacity-8 shell, the radial model lacking



The orbital picture makes the same eight as 2+6, an s pair and three p lobes. The two accounts agree on the number and differ in the decomposition, and it is the decomposition that bonding geometry sees: sp^3 hybridisation exists to turn 2+6 back into four tetrahedral directions, which 4+4 supplies at the outset.

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Figure 2: The octet as 4+4. Four unit charges on a shell settle as a tetrahedron, the arrangement that puts them as far apart as a sphere allows. A second four cannot join them — the tetrahedron is closed — but they can occupy the *dual* tetrahedron, and the eight together are the corners of a cube. The two interleave rather than stack: on a cube of side 2 the charges within one tetrahedron are 2.83 apart while the nearest charge of the other is at 2.00, so every charge's neighbours belong to the opposite four. This is why the closed octet is 4+4 and never a single shell of eight, which over-binds by 9–22% (Section 5.1); and it is the $n = 2$ case of the counting of Section 6, the $n = 1$ case being the antipodal pair.

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the angular exclusion that makes 8-on-a-shell unstable (a first signal that the closed shell is intrinsically three-dimensional). The physical packing is fixed instead by the build-up (Aufbau) of the atom, electron by electron, which is the subject of Section 5.2: the constraint that the inner shells are already filled removes the over-packed arrangements from the search, and among what remains the energy selects correctly. The capacities themselves are not fitted either — 2 and 4+4 are the $n = 1$ and $n = 2$ cases of the counting argument of Section 6.

5.2 Aufbau: how the shell structure forms

The shell structure of an atom is not a static fact to be looked up but something that *forms*, electron by electron, as the atom is built. Why the structure cannot instead be obtained by minimising over packings is worth stating precisely, because it turns out to be a structural property of the model rather than a weakness of the search.

The energy is monotone in the packing. Comparing candidate packings of the same atom, the energy decreases without interior minimum as the electrons are gathered into fewer, more compact shells. The unconstrained minimum is therefore always the most compact arrangement, in which the core is absorbed into the valence, and it is far from the reference value:

	minimum	E (Ha)	err		physical	E (Ha)	err
Li	3	-8.499	13.7%	Li	2+1	-7.444	0.45%
Be	4	-18.893	28.8%	Be	2+2	-14.607	0.41%
B	5	-35.450	43.8%	B	2+2+1	-24.020	2.6%
C	6	-59.214	56.5%	C	2+2+2	-37.420	1.1%

The physical configuration is, in each case, the *least* bound of the candidates and simultaneously the closest to the reference. A variational search over packings is thus not merely unhelpful here; it is anti-correlated with the answer.

What a build-up supplies. The failure is specific: it occurs only where the search is free to dissolve an already-formed core. An Aufbau construction never offers that. Electrons are added one at a time; the parent configuration is inherited; and a filled packing admits no more. Imposing only those three conditions — and letting the energy choose among what remains — recovers the physical configuration of Li, Be, B and C with no empirical input, the energy merely breaking ties. Nothing is frozen numerically: every candidate is solved with all shells, radii and amplitudes relaxed.

This is where the path dependence of Section 4 becomes the physics rather than a caveat. The three conditions are not restrictions imposed on a search; they are a description of the *path* the atom takes as it forms. An electron arriving at an atom whose inner shells are already occupied cannot reach them — not because a rule forbids it, but because a filled shell is the more rigid of the two. What decides is *relative* rigidity, and nothing else could: the interface moves only where one domain’s amplitude locally exceeds the other’s, so the comparison is between the two charges and not a property of either alone. The compact charge of a complete shell wins against the diffuse charge arriving at it; the newcomer is turned aside and takes the room outside instead. Figure 4 shows this happening — the core unchanged and centred while the arriving charge is deflected around it. The merged configuration remains a legitimate state of lower energy; it is simply not reachable from where the arriving charge starts. The build-up is thus not a constraint added to repair the minimisation. It *is* the minimisation, carried out along the path the atom actually takes, and the physical configuration is the equilibrium at the end of it.

For Be the two conditions already decide the outcome the energy would also choose: 2+2 lies below 2+1+1 by 0.13 eV, and 2+2 is the physical $1s^2 2s^2$. For B and C the closure condition is doing the work, since the energy alone would continue to fill the inner shell.

Where the construction is incomplete. Two limits should be stated plainly, since neither is apparent from the results above.

The build-up is exhibited as a dynamics only at its first step. Figures 3 and 4 add one charge to Li^+ and follow it, and what they illustrate is one of the two cases: the *opening of a new shell*, the arriving charge finding the inner shell closed and taking up position outside it. The other case — a charge joining a shell already open, which must then rearrange to share it — is not shown. Beryllium is its first instance, 2+1 going to 2+2 with the resident outer charge pushed into an antipodal pair, and it is the natural next step. Nothing here carries the construction further up the row. What supports Li through C is the comparison of *converged* configurations reported above — the energy

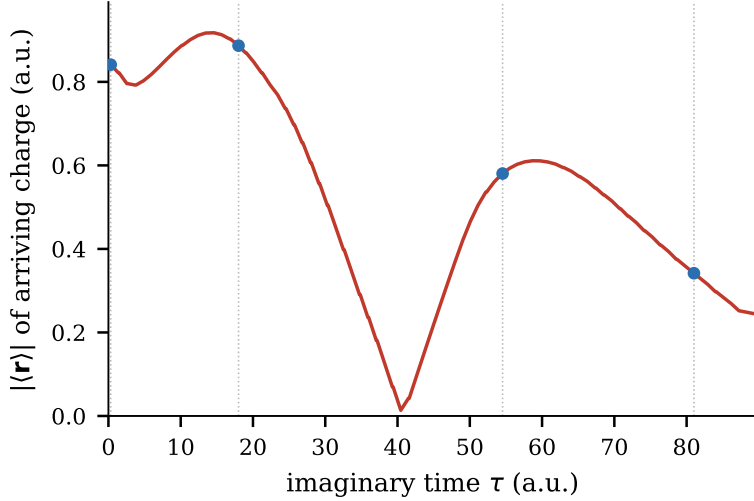


Figure 3: The build-up as a dynamics: Li^+ with one electron added. The arriving charge is launched as a *half-shell* outside the core — its density concentrated on one side, so that no shell is present at $\tau = 0$ — and the interface between the two is free thereafter. Plotted is the distance of the arriving charge’s centroid from the kernel against imaginary time. It falls from 0.84 to 0.014 — essentially through the centre — then swings back out: a damped oscillation about the closed shell rather than a decay onto it. The charge wraps *around* the core rather than merging into it. Dotted lines mark the four times shown in Figure 4. A longer run of the same case carries the oscillation through a second pass and down to 4.5% of the launch value; the run plotted here stops partway through the second swing, which is why it ends at 0.24 rather than near zero. Merging is available to the relaxation and is the lower state — a single 3-shell lies 1.055 Ha below 2+1 — so it is declined because the path cannot reach it, not because any constraint forbids it. All-electron, no r_c , 200³; the core is under-bound at this resolution, so no energy or radius is quoted from the run and none is needed for the statement made.

selecting correctly among those a build-up would leave available — which is a statement about the outcome, not a simulation of the process. We therefore offer the dynamic Aufbau as something the model makes explorable, opened at the first step, rather than as a mechanism already carried through the table.

And from nitrogen onward even the static rule is undetermined. A shell holding two charges may be a closed antipodal pair or a half-filled tetrahedron, and nothing in the present model distinguishes them; the build-up then either opens a spurious further shell or over-fills the existing one. We report the rule as established for Li through C and open beyond, rather than extrapolate it. It follows that there is no break at $Z = 18$ in the Aufbau, which stops far earlier; $Z = 18$ bounds the *energies*, whose configurations come from the capacities and the balanced rule.

Where the capacities come from. Two accounts are available and they differ in provenance, which matters more than the arithmetic.

The first is the familiar one. If a shell divides into two half-shells, each roughly a square, and a square subdivides as $n \times n$, the capacity is $2n^2$: the sequence 2, 8, 18, 32, with each n serving two periods except the first, giving period lengths 2, 8, 8, 18, 18, 32, 32 and closures at $Z = 2, 10, 18, 36, 54, 86$. It reaches radon and reproduces every noble gas. But it should be said plainly what this count is: $n^2 = \sum_{\ell=0}^{n-1} (2\ell + 1)$ is the degeneracy of a hydrogen shell, doubled by

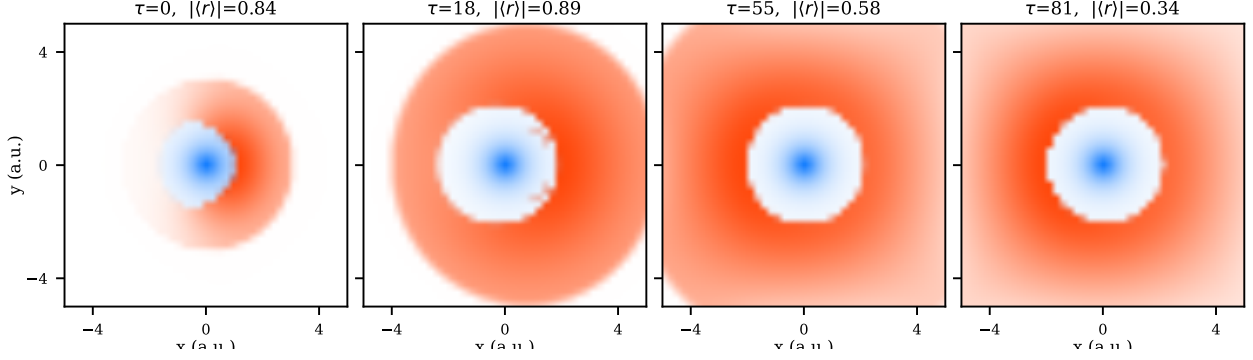


Figure 4: The same run as Figure 3, seen in space: sections through the kernel at four times, with the Li^+ core in blue and the arriving charge in red, intensity following density on its own territory. At $\tau = 6$ the arriving charge is a half-shell on one side and there is no shell anywhere; it then sweeps around the core and closes into one. The core is unchanged throughout — centred, compact, holding exactly its two charges at $r_{50} = 0.66$ a.u. — and at no time does the arriving charge enter it. This is what the non-overlap condition does in practice: the interface moves only where a domain’s amplitude locally wins, and against a formed core it does not, so the only direction available to the incoming charge is around.

spin. Offering $2n^2$ as RealQM’s own derivation of the capacities would be borrowing the answer from the theory being compared against. It also asks for 3×3 and 4×4 tilings, and the packing has no way to build them — a 3×3 tiling is not assembled from 2×2 blocks.

The second account uses nothing but what this model has measured. The basic filled units are the *antipodal pair*, 2, and the *dual tetrahedra*, $8 = 4+4$, each established above and neither imported. Composing rows from those two:

row	composition	length	closes at
1	2	2	2
2	4+4	8	10
3	4+4	8	18
4	$2 + (4+4) + (4+4)$	18	36
5	$2 + (4+4) + (4+4)$	18	54

The first five row lengths and all four closures through xenon follow from the two packings alone, with no appeal to an orbital count and no tile the model cannot build. On this reading 18 is not $9+9$ but $2+8+8$ — a pair together with two octets — which is also how the fourth row divides in fact, two elements then sixteen.

The octet rule is then doing more work than it is usually given. It is not merely the rule for the second and third rows: it is the unit out of which the fourth and fifth are built as well, 18 being a pair and two octets rather than a capacity of its own. Four row lengths, 2, 8, 8, 18, from one packing and one pair — and argon itself is the same composition read cumulatively, its eighteen charges arranged as $2+(4+4) + (4+4)$, which is what the solver returns for it without being told.

The disagreement with the orbital account is then the same at both row lengths, and it is a disagreement about decomposition rather than about number. Where that account makes the octet $2+6$, an s pair and three p lobes, this one makes it $4+4$; where it makes the fourth row $2+10+6$, a pair with five d and three p , this one makes it $2+8+8$, a pair and two octets. Both give 8 and both give 18. What differs is what the eight and the eighteen are *made of*, and that is what bonding geometry reads.

Table 1: Total atomic energies (balanced cap-4 packing, spherically-reduced solver) versus near-exact nonrelativistic references. Mean $|\text{error}| = 1.7\%$, maximum 4.6%; most atoms within 2.5%.

Z	atom	config	TE RealQM (Ha)	reference (Ha)	error
2	He	2	-2.845	-2.904	-2.0%
3	Li	2+1	-7.434	-7.478	-0.6%
4	Be	2+2	-14.630	-14.667	-0.3%
5	B	2+2+1	-24.025	-24.654	-2.6%
6	C	2+2+2	-37.445	-37.845	-1.1%
7	N	2+3+2	-55.263	-54.589	+1.2%
8	O	2+3+3	-75.389	-75.067	+0.4%
9	F	2+4+3	-102.494	-99.734	+2.8%
10	Ne	2+4+4	-131.833	-128.938	+2.2%
11	Na	2+4+4+1	-165.360	-162.255	+1.9%
12	Mg	2+4+4+1+1	-201.704	-200.053	+0.8%
13	Al	2+4+4+3	-248.671	-242.346	+2.6%
14	Si	2+4+4+4	-302.531	-289.359	+4.6%
15	P	2+4+4+3+2	-347.535	-341.259	+1.8%
16	S	2+4+4+3+3	-415.040	-398.110	+4.3%
17	Cl	2+4+4+4+3	-466.118	-460.148	+1.3%
18	Ar	2+4+4+4+4	-528.000	-527.540	+0.1%

It runs out at the sixth row. Thirty-two is not 2+8+8 repeated ($54 + 18 = 72$, not 86), and composing it from the same units requires either four octets with no leading pair or a unit the scheme does not contain. We therefore report the packing account as exact through xenon and open beyond, and we do not claim $2n^2$ as a result of this model. Which of the two is the right reading of the fourth row — 9+9 or (2+8)+8 — is not settled here; it is worth recording that they are distinguishable in principle, since only the second is expressible in the packings this model possesses — and that the energy does not distinguish them in practice. Composing krypton as the third row’s structure twice over, 2+4+4+4+4 repeated, against the same 36 charges packed as 2+4+4+4+4 | 4+3+3+2+3+3, the two differ by 1.6 Ha in 2662, or 0.06%; for zinc the corresponding difference is 0.04%. Both are a hundred times smaller than either configuration’s distance from the reference, so the comparison selects nothing. That is the same finding as Section 5.2’s, one row further up: the energy is not what fixes the structure, and two arrangements of the same charges can sit within a twentieth of a per cent of each other.

5.3 All-electron total energies, periods 1–3

Table 1 reports total energies with the balanced cap-4 packing against near-exact nonrelativistic reference values [6].

The mean relative error is 1.7% and the maximum 4.6% (Si, S); most atoms fall within 2.5%. This is comparable to Hartree–Fock accuracy on total energy — with the opposite sign of error (RealQM over-binds the heavier atoms of the third period, where HF under-binds through missing correlation) — obtained from a 1D radial solve with no basis sets, no antisymmetry, and no two-electron integrals.

5.4 Boundary of the spherical model: where radial nesting fails

The reach of the parameter-free calculation is set by the shape of the table itself. The first three rows have lengths 2, 8, 8, and the reduction covers them exactly: the packings 2, 4+4, 4+4 nest, one shell inside the next, and the radial description is the whole of the arrangement. The fourth row is 18 long. What lengthens it is the entry of the transition elements, and with them a filling that is no longer a simple outward sequence — the added charges take the *inner* position, beneath the pair that completes the row above. That placement, and not the reach of the calculation, is what bounds the *build-up*: the energies run on past argon, parameter-free, once the mesh is refined.

It is worth being exact about what fails, because it is not the geometry. The arrangement beyond argon remains a nested sequence of radial shells: the *3d* charges sit at a smaller radius than the *4s* pair outside them, which is an ordinary nesting and one the radial solver represents without difficulty — given that configuration it returns the observed removal order across the series (Section 8). What fails is the *order of filling*. Through $Z = 20$ each added charge goes outside what is already there, and a build-up that adds outward reaches the right configuration. At $Z \approx 21$ the charge added belongs to a shell *inside* the one already occupied — shown model-independently by the ionisation order, since the *4s* electrons are the first removed and are therefore the outer ones. A build-up that only adds outward has no such move, so it opens a further outer shell instead.

The distinction matters for what may be claimed, and two boundaries must be kept apart. The *energies* of Section 5.3 run to $Z = 18$, computed all-electron with no fitted length; but the configurations they are computed in come from the balanced cap-4 rule together with the capacities of Section 6, not from a build-up carried out element by element. The *Aufbau* proper — the configuration arrived at by adding one charge at a time — is demonstrated only at its first step and established only through carbon (Section 5.2). There is accordingly no sharp break at $Z = 18$ in the build-up; $Z = 18$ is where the arrangement stops being reachable outward, which is a different statement and a weaker one.

Nor is $Z = 18$ a computational boundary. On a uniform mesh the innermost shell contracts as $1/Z$ — measured here as $r_{1s}Z \approx 4.3$, constant to $\pm 9\%$ from carbon to argon — so at fixed N it is eventually spanned by too few cells and the energies degrade, turning positive around iron. That is resolution and not physics, and refining the mesh removes it: at $N = 800$ the all-electron total energies run from argon to zinc within 1–5%, with no trend in Z , and nothing is fitted anywhere. The relaxation step is explicit, $\Delta\tau = \frac{1}{2}h^2$, so the step count must be scaled as h^{-2} alongside; a refinement taken without that returns a *less* converged answer and is easily mistaken for a mesh effect.

The reader can see all of this directly. The Atom Simulator runs the whole table in a browser, at either mesh, and needs no installation:

https://claes542.github.io/RealMolecule/atom_simulator.html

We would encourage the reader to open it alongside this section. Setting Z through the first three rows reproduces the configurations of Section 5.3. Continuing past argon is more instructive still, because *nothing appears to go wrong*: the solver goes on producing neatly nested shells and converging. Scandium comes out as 2+4+4+4+4+1+1+1 — having no move that puts a charge inside an existing shell, the build-up opens a further outer shell to hold one, and then another. Physically scandium is $[\text{Ar}] 3d^1 4s^2$, one of those three charges lying *inside* the other two.

The distinction worth drawing is between representing that arrangement and reaching it. The model represents it without difficulty: told to place the *d* charges inside and the *s* pair outside, it returns the observed removal order for the whole series, which is the calculation of Section 8 and the reason we there call the placement imposed rather than derived. What the model cannot do is

arrive at it. The build-up adds outward, so it never offers the inner position, and it fails silently — converging, reporting nothing amiss, and answering the wrong question accurately. Only the reference energies and the removal order reveal it.

5.5 Electronegativity from the shell energies

The per-electron energies make one further quantity immediate. Allen’s configuration energy — the mean binding energy of the valence electrons,

$$\chi = \frac{n_s \varepsilon_s + n_p \varepsilon_p}{n_s + n_p}, \quad (4)$$

written here as the occupancy-weighted mean over the valence shells, since the model has no s/p distinction — is a state property of the neutral atom. It needs no ion, no anion, and no calibration, and it is the quantity that bond polarity is built from. The values below use the all-electron balanced packings with no r_c anywhere.

	Li	Be	B	C	N	O	F	Ne
χ this work (eV)	7.26	10.18	14.49	20.68	28.81	36.69	47.93	58.39
χ Allen (eV)	5.39	9.32	12.13	15.05	18.13	21.36	24.80	28.31
ratio	1.35	1.09	1.19	1.37	1.59	1.72	1.93	2.06

The *ordering* is reproduced exactly, element for element, across the period: the model places F above O above N above C, and so on down to Li. The *gradient* is too steep by a factor 2.23 (span 51.1 against 22.9 eV), and the discrepancy is not a constant scale — the ratio climbs monotonically from 1.09 at Be to 2.06 at Ne.

The error is thus concentrated at the electronegative end, and its cause is the same one identified in Section 5.4. Radial nesting splits the valence into an inner shell bound too tightly and an outer shell too loosely: in nitrogen the middle shell comes out at 44.3 eV where the physical $2s$ lies near 25 eV, and with three of the five valence electrons in it, it dominates the mean. The same defect appears with the opposite sign in the outermost shell. One structural cause — the outer tetrahedron placed at too large a radius — therefore produces both the steep χ gradient and the light p -block limitation, and both are removed only by a solve that keeps the angular arrangement, that is by the three-dimensional form.

The result is accordingly strongest where the model is: at the electropositive end, with Be and B within 9 and 19%. As a predictor of bond polarity it is the differences across a row that matter, and those inherit the same overestimate.

6 Discussion: packing versus orbitals

The shells s, p, d, f that organise the conventional periodic table are the *eigenmodes of the one-electron hydrogen atom* — the angular-momentum eigenstates $R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$ of $-\frac{1}{2}\nabla^2 - 1/r$, their capacities $2(2\ell + 1) = 2, 6, 10, 14$ merely counting the spherical harmonics at each ℓ (times spin). This is a single-electron, non-interacting spectrum, exact for hydrogen and for nothing else; the ℓ -degeneracy is the special symmetry of the pure $1/r$ potential, already lifted by screening in every many-electron atom. The shells enter the periodic table by *imposition*: the orbital approximation places each electron in a hydrogenic mode, fills them by the Aufbau order, and enforces exclusion through the antisymmetry of a Slater determinant.

How fully does this *explain* the periodic table? The rigorous content is single-electron: the s, p, d, f capacities $2(2\ell + 1)$ are the spherical-harmonic counts of the one-electron hydrogen spectrum, and the magic numbers $2n^2$ their sums. The many-electron structure, however, rests on rules that are *descriptive* rather than derived — the Aufbau/Madelung $(n + \ell)$ filling order and Hund’s rules are empirical regularities, not consequences of the Schrödinger equation, and the $4s$ -before- $3d$ ordering in particular has no first-principles derivation within the orbital picture. So standard quantum mechanics *reproduces* the table with great economy, but its physical explanation is layered: exact angular momentum for one electron, plus fitted filling rules for many. RealQM relocates the capacities to geometry: 2 and 4 and the octet $4+4$ follow from close-packing under Coulomb repulsion, and composing rows from those units gives the lengths 2, 8, 8, 18, 18 (Section 5.2). The *order* is then partly recovered rather than fitted — a build-up that inherits the parent configuration and closes a filled packing selects the physical configuration of Li through C with no empirical input, the energy only breaking ties (Section 5.2) — though it is undetermined from nitrogen onward, and the doubling of each capacity across two periods is not derived here. So both frameworks explain the *capacities* from genuine physics (harmonic counting; close-packing), and both still lean on an empirical element for part of the *order*: the orbital picture on the Madelung rule throughout, the present one on an undetermined choice beyond carbon.

A concrete difference is worth noting, because the two pictures decompose the *same* number differently. The octet is 8 in both, but the orbital picture splits it as $2 + 6$ (an s pair and a p sextet, by angular momentum) while the packing splits it as $4 + 4$ (two tetrahedra, by geometry). These are not notational variants: a single 8-shell over-binds by 9–22% here, so the model asserts that the physical division of the octet is the geometric one. The first shell is special in both readings for the same reason — it admits no subdivision. In the orbital picture $n = 1$ carries only $\ell = 0$, so there is no second subshell to interleave with; in the packing picture $n = 1$ is the minimal tiling, one cell per half-shell, the antipodal pair. That is consistent with its being the one capacity in the table that is not repeated.

RealQM proceeds differently. There are no eigenmodes: the electrons are localised, non-overlapping charge clouds, and the ground state minimises the Coulomb energy under spatial exclusion. What emerges is geometric *packing* — the close-packing of mutually repelling charges on a shell (the Thomson problem), whose small-number optima are the antipodal pair (cap-2) and the tetrahedron (cap-4), the octet realised as $4+4$. The two ways of realising exclusion, set out in Section 1, have different consequences for shape: the angular eigenmodes — the p dumbbells, the d cloverleaves — follow from antisymmetry, the tetrahedra from non-overlap. We make no claim as to which account is the more fundamental. What can be said is that the second admits a directly physical reading: charges that cannot occupy the same place, arranging themselves as far apart as the available room allows.

Spin illustrates the same point. In the orbital picture the factor 2 in $2n^2$ is spin, two electrons sharing one spatial function; here it is the two half-shells, since non-overlap forbids two electrons from occupying the same region and a pair must therefore be spatially divided — the cap-2 core being the antipodal counterpart of a filled s orbital. The electron count comes out the same by a different route. Spin plays no role in the present model, which is a real limitation and not merely an omission: singlet–triplet splitting, Hund’s rules, and spin magnetic moments lie outside it entirely (Section 8). We note, without demonstrating it, that the *spatial* content of Hund’s first rule — that electrons keep apart — is enforced here unconditionally by non-overlap, whereas its energetic content is absent.

A regularity of the packing, borne out by the table, sharpens the point: **every period closes as $4+4$, never as a single 4**. The two are distinct objects. A single tetrahedron has T_d symmetry and is *directional* — four vertices, no inversion centre — whereas $4+4$ places the second tetrahedron

as the *dual* of the first, the pair forming the cube (O_h symmetry, the eight vertices of the *stella octangula*). A closed, inert shell must be spherically symmetric, and only the cube achieves this; the single tetrahedron is its directional, *reactive* half. The octet rule is then geometric — an atom closes its directional 4 into the spherical 4+4 — and carbon’s tetravalence is precisely the four bond directions of its single tetrahedron, completed to the cube by sharing one electron with each of four partners. (Equivalently, 8 is two *dual* tetrahedra forming one cubic shell, of which the radial 4+4 is the 1D shadow — which is why the radial model places the outer tetrahedron too far out, the limitation of Section 5.4.)

RealQM moreover gives its *own* rationale for the capacities, with no appeal to the harmonic count. Each shell is two *dual half-shells* — the pairing realised geometrically, the two interpenetrating tetrahedra of the octet being the $n = 2$ instance — and for the octet each half is 1+3, an apex over a triangle. Whether that subdivision continues as n^2 for larger shells, and so returns $2n^2$, is the open question of Section 5.2; the packings established here are the pair and the octet. The ring counts $2\ell + 1$ are the geometric counterpart of the angular degeneracy: for $n = 2$ a half is 1+3 — an apex over a triangle, i.e. a tetrahedron — and the two duals form the cube. Packing alone thus yields both the magic numbers and their $2\ell + 1$ substructure, with no angular-momentum operator.

This also explains why 8 is privileged. Only $n = 2$ gives a half (1+3 = 4) whose double is a *Platonic* closed surface (the cube); for $n \geq 3$ the half (9, 16, ...) is a regular division with no unique polyhedron. So 8 is a closed, spherically-symmetric *surface*, whereas 18 and 32 are *nested composites* — a cube wrapping interior *d*- and *f*-packings. Three facts follow at once: every period closes as the octet 4+4 (the only available closed surface), the *d*- and *f*-blocks are *buried* (no surface carries them), and the long periods are octet surfaces over interior shells, not larger noble-gas surfaces.

The two pictures agree on the magic numbers 2, 8, 18, 32 because both count arrangements on nested spheres in three dimensions — the same geometry read spectrally (the harmonic count $2n^2$) or geometrically (the packing capacity) — but they decompose them differently: 2+6+10+14 into sub-shells versus 4+4, 4+4+2, 4+4+4+2 into tetrahedra and pairs. A closed shell is spherically symmetric either way; the pictures part company in the angular correlation of open shells and, decisively, in *bonding geometry*. There the geometric picture is the one chemistry already uses: VSEPR and sp^3 hybridisation are geometric repairs grafted onto the orbital model precisely because the bare eigenmodes do not give molecular shape — one must *hybridise* *s* and *p* into four tetrahedral lobes to recover what the cap-4 packing yields directly. RealQM does not claim the orbitals are wrong; it locates them as the *spectral shadow of the one-electron problem*, while the structure of the real interacting atom — and of the chemistry built on it — is geometric packing.

7 Numerical properties of the solver

Spatial discretisation. We measure the total energy under grid refinement at fixed box size, estimating the observed order p from $|E_h - E_{h/2}| \sim h^p$ (Table 2). For He the convergence is clean and **first order** ($p = 0.94$): the nuclear cusp — a kink in u at $r = 0$ — caps the formally second-order stencil at $\mathcal{O}(h)$, which is also why the light-atom total energies retain a ~ 1 –2% residual at the production grid.

The inner $1s$ contracts as $1/Z$, so a uniform grid must be refined in proportion to resolve it: at the coarse N above the Ne ($1s$ at $r \approx 0.1$) and Ar ($r \approx 0.056$) cores are unresolved and the energies diverge. This $1/Z$ resolution requirement is the numerical face of the radial-nesting boundary (Section 5.4): it is a second reason, independent of the packing, why the all-electron table stops at argon.

Table 2: Grid-refinement convergence of the total energy at fixed box (He). The order is first ($p \approx 1$), cusp-limited. Heavier atoms require the grid to resolve the inner $1s$ ($r \sim 1/Z$): at this coarse N the Ne and Ar cores are under-resolved and the energies are meaningless, so the production runs use $N = 400$ (see text).

Atom	N	h (a.u.)	E_{total} (Ha)	$ E_N - E_{2N} $	order p
He	64	0.078	-2.79543	—	—
He	128	0.039	-2.82554	0.03012	—
He	256	0.020	-2.84121	0.01566	0.94

Where the error is. It is worth saying plainly which approximation dominates. The kernel is the bare $-Z/r$, and near the nucleus a uniform grid does not resolve it: the innermost sampled value is Z/h , and the $1s$ it binds spans some eight cells at argon. The consequence is measurable. Argon’s two-electron core comes out 69 Ha too deep against the He-like value, a fifth of it, while the neutral atom lands within 0.1% — so that agreement is a cancellation of the core error against the rest, not accuracy; and the total moves 10.7 Ha between $N = 400$ and $N = 800$, the relaxation being converged at both. In the light atoms the same effect appears as first-order behaviour, doubling when h doubles. Beside this the packing, the free boundaries and the Coulomb sums are second-order effects: the unresolved $1/r$ singularity is the largest single contribution to the errors reported here.

What convergence means in three dimensions. The ladder above belongs to the radial solver, where a total energy is quoted and its limit is the question asked. The In the present setting — the simplest possible implementation, on the smallest systems — the three-dimensional calculations display *qualitative* features of the solution rather than precise energies or geometry: which charge holds which region, where an interface stands, whether an arriving charge closes into a shell rather than merging inward. Figures 1 to 4 are of that kind, and are reported as such. What is asked of refinement there is that these features do not change under it, the mesh representing shape rather than serving as a quadrature. The precise numbers in this paper come from the radial solver, where the ladder above applies.

Iteration and free-boundary convergence. The imaginary-time residual $\|\dot{u}_s\|$ and the boundary jumps $|u_s(R_s) - u_{s+1}(R_s)|$ both decay to steady state; at convergence the jumps vanish, verifying the Bernoulli (continuity) condition at every interface.

Throughput. Each step is an $\mathcal{O}(N)$ one-dimensional grid sweep over the radial mesh; a single atom converges in a fraction of a second and the whole of periods 1–3 in seconds on a laptop CPU — no basis sets, no two-electron integrals, no N -body configuration space.

8 Limitations

Spin. The model contains no spin. Exclusion is realised geometrically, by non-overlap of domains, and the electron counts that spin supplies in the orbital picture are here supplied by the two half-shells (Section 6). This is a limitation of substance, not of presentation: singlet–triplet splitting and exchange energy as such, Hund’s rules and multiplicity, spin magnetic moments, fine structure, Zeeman and ESR responses, and the ortho/para distinction in helium all lie outside the model as

formulated. Nothing reported here depends on them — ground-state energies and shell structure do not — but any extension to magnetic properties or to multiplet spectra would require the model to be enlarged, not merely refined.

Only the stationary (converged) solution is used, so the temporal order of the imaginary-time relaxation is immaterial. The relevant limitation is the *spatial* discretisation: it is low order and cusp-limited (first order at the nucleus, Section 7), so the light-atom total energies retain a $\sim 1\text{--}2\%$ residual that a higher-order stencil or a logarithmic radial mesh would reduce. For the transition metals the *d*-shell must be placed *inside* the *s*-shell for the arrangement to be consistent at all; that placement is imposed, not derived, and we do not claim the ordering as a result. The interpenetrating *4s/3d* structure is not sorted out here. Open-shell and strongly non-spherical configurations test the angular representation and are flagged where relevant; electron affinities are a known difficult case and lie outside this paper.

9 Conclusion

A periodic table has been computed from scratch: the ground states of the neutral atoms, every element solved the same way, with nothing fitted.

Total energies come out within one to two percent across the first three periods, and the interface radii out of the minimisation. The result organises into one statement: packing forms the structure, the Aufbau selects which packing forms, and the energy realises it — none of the three suffices alone.

Two things follow from geometry alone and need no parameter: the octet as 4+4, two dual tetrahedra forming a cube, which a single shell of eight over-binds; and, composing rows from that octet and the antipodal pair, the lengths 2, 8, 8, 18, 18 with closures at the noble gases through xenon. The electronegativity ordering across period 2 follows from the shell energies directly.

The reach of the calculation is set by the same geometry. The first three rows have a clear structure — shells nesting simply, capacities 2, 8, 8 — and there the spherically reduced form captures the essence of the table, an atom being spherically symmetric. From the fourth row the shells interpenetrate, the *3d* lying inside the *4s* that fills after it; that structure we have not sorted out in detail, and the reduction fails there. The three-dimensional form of the model carries no such restriction, and it is the form that extends to molecules.

10 Code and reproducibility

The three relaxations of Section 4 are the whole algorithm, so the code implementing them is the simplest an atom admits. Two forms are used:

	scope	solver
spherically symmetric (1D radial)	all Z	≈ 175 lines
Cartesian (3D)	small Z , no reduction	≈ 300 lines

The counts are of the solver proper — the imaginary time stepping, the Poisson solve and the boundary move, with their initialisation. Each source file carries at its head the three updates *quoted in their own code form*, with the line at which each appears, so that a reader can check the three-line claim against the lines that run rather than against a paraphrase of them. We would make a general point of this. A model whose content is three update rules is more honestly presented by exhibiting those rules as executable text than by transcribing them into display equations and asking the reader to trust that the program implements what is written: the code is where the

approximations, the boundary handling and the discretisation actually live, and a short program is a stricter specification than a long derivation. Python or JavaScript of this size is, we think, a reasonable medium for computational physics, and the appropriate object of review alongside the text.

The contrast with established practice is worth stating, and not as disparagement. The standard quantum-chemistry and electronic-structure packages run to hundreds of thousands or millions of lines, written by large teams of computational physicists over decades and without AI assistance of any kind, and they are mature and heavily validated. But no one reader holds such a code in view. In practice its physics is taken on the authority of the community that maintains it rather than read, and the distinction between what the model asserts and what the implementation does is not available for inspection. The codes here are short for a different reason, and not because anything has been conceded to get them short. They are short because the model is nothing but the physics: charge densities in space, the Coulomb interaction between them, and the three updates that carry them to equilibrium. No computational apparatus is laid over the physics to make it tractable, and the length of the code follows from that. Nothing is given up in reach — the same three updates run from atoms to molecules to condensed phases — nor in speed, the work scaling as N times the mesh and so linearly in the number of electrons. Nor, measured as accuracy per unit of computation, in accuracy: the whole table to one or two per cent in about a minute on a laptop. What the size additionally buys is that the entire content of the calculation can be read by one person in an afternoon — which for a paper claiming that a model reproduces a periodic table with nothing fitted is not a small thing, since the claim is worth no more than a reader’s ability to confirm it.

There is a practical side to this. The codes here were developed with AI assistance from p5.js templates written by the author, and those templates are essentially the three update lines displayed above: the content of the method was authored directly, and what assistance supplied is the scaffolding around it — initialisation, plotting, batch drivers — none of which enters a result. The reasonable worry about AI-assisted work is that the reasoning behind a program becomes unavailable for inspection, that the code turns into the part of a paper nobody reads. A program short enough to be read in full is the answer to that worry rather than an instance of it: what was written is on the page, in the form a referee reads it, and can be checked line by line against the physics it claims to implement. Not all coding should be hidden by AI, and least of all the few lines that carry the content of a model. The enclosing pages add plotting, a user interface and the batch harnesses that produce the tables, and are larger; nothing in them enters the computation. Neither solver uses a basis set, a library eigensolver, or a fitted functional, and both are short enough to be read end to end. That is the point of reporting their size: the claim that no parameter is adjusted anywhere is one a reader can check directly rather than take on trust.

The molecular and condensed-phase work uses larger implementations of the same three relaxations — a labelled-domain solver and a GPU solver, a few thousand lines each, which reach systems of thousands of atoms. Neither is used for any result reported here; both are described with the work in which they are applied [1].

The complete implementation, the validation suite, and an interactive gallery are publicly available:

- Source code: <https://github.com/Claes542/RealMolecule>
- Interactive gallery: <https://claes542.github.io/RealMolecule/>
- Atom Simulator, the spherically reduced solver, running in a browser: https://claes542.github.io/RealMolecule/atom_simulator.html

The last of these is the code whose three updates are quoted above, with a live display of the shells, their free boundaries and the energies as the relaxation proceeds. A reader who wants to know what the model does, rather than what we say it does, will learn more in ten minutes there than from this section. Every figure and table is reproducible from the published source on the hardware stated in Section 7.

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